Nokia multispan dataset description

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The Nokia multispan system data are saved in a custom binary file format, with a new file being made every time the buffer fills up. This means that the shape of data arrays varies over the experiment, and each file does not correspond to a given amount of time. For instance, for a time when the channel spacing was large and a spatial mask was applied, the data file will contain more time points, and there will be a larger space in time between collected files. In contrast, when spatial sampling is high, there will be more overall files for a given day, and each file will contain fewer points along the time axis.

The data directories are organized by time and cable as follows. The subdirectories with 100 m channel spacing only appear after December 23 2025.

Year
→ Month
→ → Day
→ → → North cable
→ → → → 8 Hz
→ → → →→ *binary data files*
→ → → South cable
→ → → → 8 Hz
→ → → →→ *binary data files*
→ → → North cable, 100 m channel spacing
→ → → → 8 Hz
→ → → →→ *binary data files*
→ → → South cable, 100 m channel spacing
→ → → → 8 Hz
→ → → →→ *binary data files*

The files are named as: *experiment_cable_samplingrate_date_time*

The time listed in the filename is related to the time that the previous file was written, not the start time of the data within the file. For example, the times listed in the filenames in one directory are:

1. 20251118_203019
2. 20251118_203020
3. 20251118_204706

But the start times of the data within the files are as follows:

1. 2025-11-18T20:30:20.765052Z
2. 2025-11-18T20:47:06.868339Z
3. 2025-11-18T21:03:52.968312Z

However, during some data outages and parameter changes, the time in the file name does not always follow this pattern, depending on how the data writing was restarted. The timing of the data should always be read from the metadata within the file itself.

In data post-processing, it was found that due to latency in the processing chain, **the absolute timestamps associated with the data are delayed (behind real-time) by several seconds**. This offset cannot be calculated from available metadata, and it changes throughout the experiment with changes in acquisition parameters. This will be addressed in future experiments. Note that this timing issue is an absolute offset; the relative timestamps of each sample are correct. See details in the “Timing Offset” section for more explanation.

Sampling rate and channel spacing were varied throughout the experiment. The dates of changes, alongside the times of data outages, are summarized below.

Date	**Sampling Rate**	**Channel Spacing**
Nov 5*	29.8 Hz	105.9 m
System stops recording on November 12 at ~20:00 UTC		
System restarts recording on November 18 at ~20:30 UTC		
Nov 18	29.8 Hz	105.9 m
Nov 26	7.45 Hz	52.9 m
Files become corrupt/cut out Dec 8 ~17:00 UTC		
Data comes back online with valid files on Dec 9 ~22:00 UTC		
Dec 9	7.45 Hz	52.9 m
Power outage, data cuts out around Dec 17 ~07:00 UTC		
Data comes back online Dec 23 ~19:00 UTC		
Data is now available in two portions: with 100 m spacing and otherwise		
Dec 23	1.86 Hz, 7.45 Hz	105.9 m, 13.2 m
Dec 30	1.86 Hz, 7.45 Hz	105.9 m, 26.5 m
Jan 6	1.86 Hz, 7.45 Hz	105.9 m, 211.7 m
Jan 13	3.73 Hz, 14.9 Hz	105.9, 211.7 m
Jan 21	1.86 Hz, 14.9 Hz	105.9 m, 211.7 m

Jan 28	end of experiment	

*The first part of Nov 5 involved lots of parameter changes during testing, and metadata may not always reflect what is in the files at this point in time. The parameters listed here are correct for the end of the day (UTC) on Nov 5.

Data for these times were collected at rates higher than the Navy-limited 8 Hz rate, and needed to be manually downsampled to 7.45 Hz after collection. These files are indicated with "downsampled_" as a file prefix and are in h5 format rather than binary. The metadata within these h5 files indicates their new sampling rate. Information on how to read these files can be found in the multispan notebook in the Github listed below.

An example of how to read the data files using python can be found at:
https://github.com/uwfiberlab/OOI_DAS_2025/

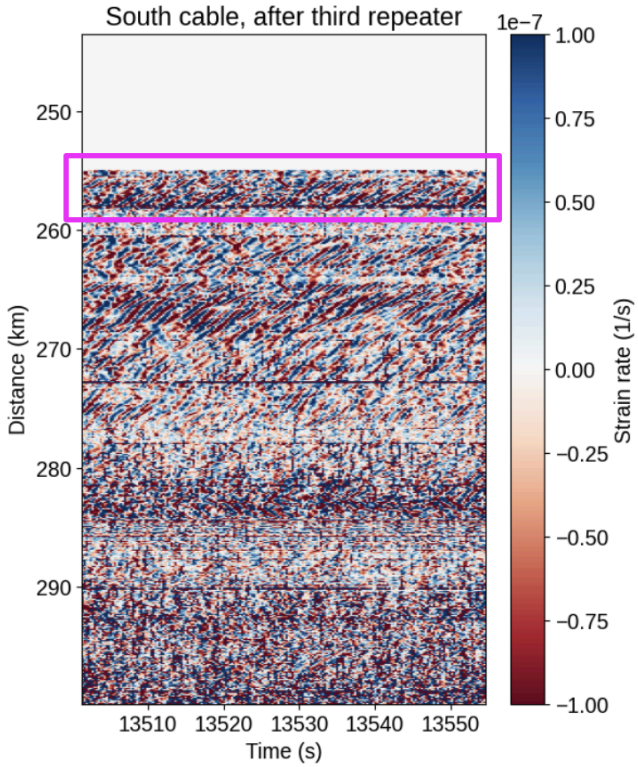
Notebooks within this repository also demonstrate how the spatial mask (applied after December 2) can be used to estimate channel locations along the cable. The cable route that can be used to interpolate for channel locations can be found <wherever we decide to link it>. The notebook also demonstrates:

- How to calculate sampling rate and channel spacing from file metadata
- How to convert the raw phase data to strain rate data

Cable/observatory nonlinearities

As mentioned in the experiment description, there are significant nonlinearities within the OOI fibers in addition to an observatory-related signal on the receive fiber of both cables that interferes with DAS data collection. These combined effects mean that significant portions of the cable did not record physically accurate data, which prompted the introduction of the spatial mask. However, the spatially masked data still contains some non-physical noise over significant cable portions, and should be treated with caution.

For example, below shows a waterfall plot of ~60 s of background noise (assumedly mostly ocean waves) on the south cable for the unmasked portion past the third repeater (~255 km along the cable). Visually, the data contains some along-cable portions that appear to be mostly nonphysical noise (i.e., 258-260 km, 290+ km), but other portions that appear to have ocean wave signal (i.e., 261-263 km, 265-268 km). The current recommendation is to use the first portion of "good" data past the repeater (highlighted here in magenta). **Further investigation into the characteristics of this noise are ongoing and updated recommendations will be made here moving forward.**



Timing offset

The absolute timestamps associated with the data (defined by stream.proc_start in the chunk_meta of the data files) are delayed by several seconds behind real time. The magnitude of this offset changes with the sampling rate, i.e., the packet size. Using relative earthquake arrivals between the OptoDAS dataset and ocean bottom seismometers on the south cable, we visually estimate this timing delay to be ~7 s when the sampling rate is 7.45 Hz (example from the M4 earthquake on November 5), and ~12-13 s when the sampling rate is 3.73 Hz (example from the M6 earthquake on January 16). We suggest that users manually align their data to expected arrivals in post-processing based on comparison to correctly timestamped data streams (i.e., the OptoDAS and cabled ocean bottom seismometers), or simply work in a relative time frame rather than absolute.

This timing latency is completely deterministic, so once it is estimated for a given set of acquisition parameters, it will always be the same for that stream relative to the stream.proc_start in the metadata.